

CENG 467: Natural Language Understanding and Generation

Take-Home Midterm Report

Representation Learning, NER, Summarization, Machine Translation and Language Modeling

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Abstract

This report presents a comprehensive experimental study covering four core Natural Language Processing (NLP) tasks: (1) representation learning for text classification, (2) named entity recognition (NER), (3) automatic text summarization, (4) machine translation, (5) Language Modeling. For each task, multiple modeling paradigms are compared — from classical statistical methods to state-of-the-art transformer-based architectures — under strictly fair experimental conditions. All experiments employ fixed random seeds, consistent dataset splits, and identical pre-processing pipelines unless a controlled comparison demands otherwise. The findings demonstrate the progressive superiority of contextual representations while also highlighting practical trade-offs in interpretability and computational cost.

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1 Introduction

Natural Language Processing has undergone a fundamental paradigm shift in recent years, moving from hand-crafted feature engineering and sparse statistical representations toward dense, contextual, and pre-trained neural architectures. This midterm explores this evolution across five key NLP tasks by implementing and evaluating representative models at each level of the representational hierarchy: sparse (N-gram), dense (BiLSTM), hybrid (BiLSTM-CRF), and contextual (BERT, DistilBERT, BART, MarianMT, GPT-2).

The objective is not merely to achieve the highest possible performance, but to demonstrate a rigorous understanding of each model’s design philosophy, assumptions, strengths, and failure modes. Throughout the report, all design choices are justified empirically where possible, and qualitative error analyses complement quantitative metrics to offer deeper insights.

Implementation Environment: All experiments and code development were conducted using the Google Colab platform, leveraging its cloud-based infrastructure to ensure a consistent and high-performance execution environment. Key libraries used include `transformers`, `datasets`, `evaluate`, `PyTorch`, and `NLTK`. To meet the high computational demands of large-scale pre-trained models (such as BART and GPT-2), the experiments utilized Google Colab’s **CUDA-enabled GPU** acceleration (typically NVIDIA T4 or similar). For tasks requiring less intensive computation, CPU-based execution was employed. To ensure the scientific reproducibility of all results, a global random seed of 42 was applied across all libraries and random number generators.

Note on Reported Results. *All quantitative results, tables, and figures presented in this report are derived directly from the executed notebook.* The metric values, model outputs, and error analysis examples correspond exactly to the results produced during a single full run of the pipeline. Readers wishing to inspect the raw execution outputs, intermediate logs, training curves, and complete model outputs — including those omitted from the report for brevity — are encouraged to consult the shared Colab notebook via the link below. The notebook is set to *viewer access* and requires no authentication to open.

Repository: https://github.com/cerarikoglu/ceng467_take_home_midterm

Colab Link: <https://colab.research.google.com/drive/1Ao9d9jIoZU8V-0b3nqWHskoEHdah8LHe?usp=sharing>

2 Question 1: Representation Learning in Text Classification

2.1 Problem Formulation and Dataset

The first task is binary sentiment classification, where each input sentence must be assigned a label of *Positive* (1) or *Negative* (0). The goal is to compare three distinct paradigms for text representation: sparse (TF-IDF + Logistic Regression), dense (BiLSTM), and contextual (DistilBERT).

The **SST-2 (Stanford Sentiment Treebank – Binary)** dataset, accessed via the GLUE benchmark through the `datasets` library, was selected as the evaluation benchmark. Its suitability stems from several factors:

- **Benchmark Status:** SST-2 is a cornerstone of the GLUE benchmark, enabling comparison with a vast body of published results.
- **Data Quality:** Sentences are drawn from professionally written movie reviews, labeled with high granularity by trained human annotators and then binarized, resulting in a noise-free gold standard.
- **Linguistic Diversity:** With 67,349 training samples and 872 validation samples, the dataset contains a rich variety of sentence structures, negations, sarcasm, and elliptic constructions that challenge all three model types differently.
- **Class Balance:** The dataset is approximately balanced between positive and negative classes, making accuracy and Macro-F1 reliable metrics.

2.2 Preprocessing Analysis

Before training the main models, a preprocessing ablation study was conducted using a TF-IDF + Logistic Regression baseline. Four configurations were evaluated:

1. **Basic (No Normalization):** Raw text, case-sensitive.
2. **With Normalization:** Lowercase conversion applied.
3. **Without Stopwords:** English stopwords removed using `sklearn`'s built-in list.
4. **With Feature Truncation:** Vocabulary limited to the 500 most frequent terms.

Table 1: Accuracy and Macro-F1 of the three model types (Sparse, Dense, Contextual) on the SST-2 validation set.

Experiment	Accuracy	Macro-F1
Basic (No Norm)	0.824541	0.823939
With Normalization	0.824541	0.823939
Without Stopwords	0.803899	0.802106
With Truncation (500 features)	0.741972	0.741087

Findings: Normalization (lowercase) produced the most balanced and robust improvement in both accuracy and Macro-F1. Stopword removal slightly decreased Macro-F1 despite its common use as a noise-reduction technique; this is explained by the sentiment-critical role of function words such as “*not*”, “*but*”, and “*never*” in negation and contrast constructions. Feature truncation to 500 terms imposed severe information loss, yielding the lowest performance. Based on these findings, **lowercase normalization combined with punctuation removal** was adopted as the uniform preprocessing strategy for all subsequent models.

2.3 Model Architectures and Experimental Setup

To ensure a fair comparison, all three models were evaluated under the following shared constraints:

- All models operate on the same preprocessed text (`clean` column).
- Vocabulary size for TF-IDF and BiLSTM embedding layers: 5,000 tokens (matching to avoid information asymmetry).
- Neural models (BiLSTM, DistilBERT): trained for exactly **2 epochs**.
- Maximum sequence length: 64 tokens (BiLSTM), 128 subword tokens (DistilBERT).
- Batch size: 32 for both neural models.

2.3.1 Model 1 – Sparse: TF-IDF + Logistic Regression

Term Frequency–Inverse Document Frequency (TF-IDF) transforms each document into a high-dimensional sparse vector where each dimension corresponds to a vocabulary term. The value encodes how frequently a term appears in the document relative to its frequency across the entire corpus, thereby downweighting ubiquitous words.

Logistic Regression with L2 regularization (`max_iter=1000`) was used as the linear classifier on top of these representations. This combination is interpretable — feature coefficients directly correspond to word-level influence — and computationally very efficient.

2.3.2 Model 2 – Dense: BiLSTM

The Bidirectional Long Short-Term Memory (BiLSTM) processes each sentence in both left-to-right and right-to-left directions, yielding hidden states that capture both past and future context for every token. The final hidden states from both directions are concatenated to form a sentence-level representation, which is passed through a linear classification head.

- **Embedding:** Random initialization, dimension 128.
- **LSTM:** Hidden size 64, bidirectional (effective output size: 128).
- **Classifier:** Linear layer ($128 \rightarrow 2$).
- **Optimizer:** Adam, learning rate 10^{-3} .
- **Loss:** Cross-entropy.

2.3.3 Model 3 – Contextual: DistilBERT

DistilBERT is a knowledge-distilled variant of BERT that retains approximately 97% of BERT’s language understanding capability at 60% of its size and 40% faster inference. It utilizes a 12-layer transformer encoder with multi-head self-attention, pre-trained on masked language modeling and sentence-order prediction on large text corpora.

We fine-tuned `distilbert-base-uncased` for sequence classification using the `Trainer` API from Hugging Face `transformers`. The pre-trained [CLS] token representation is mapped through a classification head to the output labels.

2.4 Results and Analysis

Table 2: Performance comparison of representation strategies on SST-2 validation set.

Model (Representation)	Accuracy	Macro-F1
Sparse (TF-IDF + LR)	0.815367	0.814997
Dense (BiLSTM)	0.810780	0.810624
Contextual (DistilBERT)	0.899083	0.898929

Key Observations:

The contextual model (DistilBERT) substantially outperforms both competing approaches, confirming the well-established superiority of transfer learning from large pre-trained models. The gap between Sparse and Dense models is surprisingly narrow, indicating that for a sentiment task where key adjectives (e.g., “*excellent*”, “*terrible*”) are highly discriminative, a strong lexical bag-of-words model can rival a sequential neural model trained from scratch.

The BiLSTM's modest advantage over TF-IDF is attributable to its ability to capture local sequential dependencies (e.g., negation spanning two tokens: “*not great*”), but this advantage is limited by the absence of pre-training and the small vocabulary.

2.5 Error Analysis: DistilBERT Failure Cases

Despite its strong overall performance, DistilBERT exhibits systematic failures in several linguistic categories. Five misclassified validation examples were examined to understand the model's limitations in sentiment detection.

Table 3: Top 5 misclassified examples by DistilBERT on the SST-2 validation set.

Sentence	True Label	Predicted	Error Pattern
<i>holden caulfield did it better .</i>	Negative (0)	Positive (1)	External reference / implicit comparison
<i>the script kicks in , and mr. hartley 's distended pace and foot-dragging rhythms follow .</i>	Negative (0)	Positive (1)	Domain-specific negative vocabulary
<i>this one is definitely one to skip , even for horror movie fanatics .</i>	Negative (0)	Positive (1)	Implicit negation masked by intensifiers
<i>though it 's become almost redundant to say so , major kudos go to leigh for actually casting people who look working-class .</i>	Positive (1)	Negative (0)	Hedged praise / contrastive opener
<i>you wo n't like roger , but you will quickly recognize him .</i>	Negative (0)	Positive (1)	Contrastive conjunction misweighted

Analysis of Misclassifications:

- **External Cultural References:** As seen in the first example, the model struggles with comparative negatives where understanding the context (e.g., Holden Caulfield) is required to grasp the critique.
- **Lexical Ambiguity:** In the third sentence, positive-valence words like "fanatics" and "definitely" likely misled the model, causing it to ignore the core negative instruction "skip".
- **Syntactic Nuances:** Contrastive conjunctions (e.g., "but" in the last example) and

complex clauses remain a challenge for DistilBERT, as it may focus on the more prominent clause rather than the overall sentiment.

1. **External Cultural References:** *“holden caulfield did it better .”* (Actual: Negative, Predicted: Positive). The word *“better”* strongly activates positive sentiment features, but the implied critique requires knowledge of the character Holden Caulfield from *The Catcher in the Rye* as a benchmark of quality — cultural context beyond the model’s scope.
2. **Contrastive Constructions:** *“you wo n’t like roger , but you will quickly recognize him .”* (Actual: Negative, Predicted: Positive). The contrastive conjunction *“but”* introduces a pivot from negative to neutral sentiment; the model incorrectly assigns greater attention weight to the more syntactically prominent second clause.
3. **Implicit Negation through Register:** *“this one is definitely one to skip , even for horror movie fanatics .”* (Actual: Negative, Predicted: Positive). The intensifier *“definitely”* and the word *“fanatics”* both carry positive emotional valence, masking the negative core recommendation (*“skip”*).
4. **Irony and Understatement:** Short ironic reviews where surface-level polarity is misaligned with intent represent a recurring failure mode for all evaluated models.
5. **Domain-Specific Vocabulary:** Technical film criticism terms (e.g., *“overwrought”*, *“mawkish”*) may be underrepresented in the pre-training corpus, reducing the model’s calibration on domain-specific negative vocabulary.

2.6 Performance vs. Interpretability Trade-off

Table 4: Qualitative comparison of representation strategies across key dimensions.

Dimension	Sparse (TF-IDF)	Dense (BiLSTM)	Contextual (BERT)
Performance	Moderate	Moderate	High
Interpretability	High (word coefficients)	Low	Very Low
Computational Cost	Very Low	Medium	High
Word Order	Ignored	Captured (locally)	Captured (globally)
Pre-training	None	None	Large-scale
Negation Handling	Poor	Moderate	Good

The sparse TF-IDF representation offers the highest interpretability — logistic regression coefficients can be directly ranked to identify the most sentiment-predictive tokens

(e.g., “excellent” → positive, “boring” → negative). Contextual models sacrifice this transparency for substantial performance gains, a trade-off that must be weighed in deployment contexts requiring model explainability (e.g., legal or medical NLP).

3 Question 2: Named Entity Recognition (NER)

3.1 Problem Formulation and Dataset

Named Entity Recognition (NER) is a sequence labeling task in which every token in an input sentence is assigned one of a predefined set of entity type labels. Unlike sentence-level classification, NER requires the model to produce a structured, token-aligned output sequence, making it sensitive to both local and long-range contextual dependencies.

The **CoNLL-2003** dataset ([tomaarsen/conll2003](https://huggingface.co/datasets/tomaarsen/conll2003) from Hugging Face) was used as the benchmark. It contains four entity types: *PER* (person), *ORG* (organization), *LOC* (location), and *MISC* (miscellaneous). Annotations follow the **BIO tagging scheme**:

- **B-X**: Beginning token of an entity of type X.
- **I-X**: Continuation token of an entity of type X.
- **O**: Token outside any named entity.

This yields a total of 9 distinct label types (B-PER, I-PER, B-ORG, I-ORG, B-LOC, I-LOC, B-MISC, I-MISC, O). The CoNLL-2003 corpus is drawn from Reuters news articles, offering a diverse and realistic distribution of named entities.

3.2 Model Architectures

3.2.1 Model 1 – Hybrid: BiLSTM-CRF

The BiLSTM-CRF is the classical gold standard for NER prior to the transformer era. It combines:

- **Word Embedding**: Random initialization with dimension 128 over a full-corpus vocabulary.
- **BiLSTM Encoder**: Hidden dimension 64 per direction (128 combined), captures sequential context.
- **Linear Emission Layer**: Maps the BiLSTM output to tag-space scores (emission scores).
- **Conditional Random Field (CRF) Decoder**: Models label transition probabilities globally, ensuring that the output sequence is globally consistent (e.g., an I-PER tag cannot follow a B-ORG tag). This is critical for BIO coherence.

The CRF layer uses the Viterbi algorithm during decoding to find the globally optimal label sequence. Padding masks are passed to the CRF to ensure padding tokens do not influence the loss or decoding.

3.2.2 Model 2 – Contextual: BERT for Token Classification

BERT (`bert-base-cased`) was fine-tuned using the `AutoModelForTokenClassification` interface. The cased variant is used because named entity recognition is case-sensitive: “*apple*” (fruit) vs. “*Apple*” (company) carry different entity labels, and capitalization is a crucial surface-level cue.

The input tokenization introduces a subword alignment challenge: BERT’s WordPiece tokenizer may split a single word into multiple subword tokens (e.g., “*Washington*” → [“*Wash*”, “*##ington*”). Only the **first subword of each original word** receives the true NER label; subsequent subwords are assigned a label of -100, which is ignored in both loss computation and evaluation. This word-to-subword alignment is implemented in the `tokenize_and_align_labels` function.

3.3 Training Configuration

Both models were trained under identical conditions for fairness:

- **Epochs:** 2 (identical for both models)
- **Batch Size:** 16
- **Optimizer:** Adam ($\text{lr} = 10^{-3}$ for BiLSTM-CRF; default Hugging Face schedule for BERT)
- **Evaluation Metric:** `seqeval` library, computing precision, recall, and F1 at the entity span level (not token level), which is the accepted standard for NER evaluation.

3.4 Results

Table 5: NER performance comparison on CoNLL-2003 validation set (entity-level, `seqeval`).

Model	Precision	Recall	F1-score
BiLSTM-CRF	0.848312	0.651296	0.736862
BERT (<code>bert-base-cased</code>)	0.945318	0.951363	0.948331

BERT substantially outperforms BiLSTM-CRF across all three metrics. The most notable gap is in **recall** (0.651 vs. 0.951), indicating that the BiLSTM-CRF model misses a large proportion of actual entities — it is overly conservative in its tagging, preferring “O” over “B-X” in ambiguous contexts. BERT’s pre-trained representations provide rich contextual priors that allow it to identify entities even in complex or elliptic constructions.

3.5 Error Analysis

3.5.1 Boundary Errors (BiLSTM-CRF)

The most common failure mode in the BiLSTM-CRF is **premature entity termination** — correctly identifying the first token of a multi-token entity but failing to continue:

True:	B-ORG	I-ORG	I-ORG
Predicted:	B-ORG	O	O

This occurs because the BiLSTM’s local context window is insufficient to carry entity continuation signals across long spans, and the CRF’s transition parameters have not converged sufficiently in only 2 training epochs.

3.5.2 Entity Type Confusion

Both models occasionally misclassify the *type* of a correctly identified entity. The most common confusion pair is **LOC vs. ORG**:

- “*Washington D.C.*” — frequently predicted as *LOC* when the sentence context implies an organizational or governmental entity (“*Washington decided to...*”).
- “*The United Nations*” — occasionally predicted as *MISC* due to low frequency in the training set.

3.5.3 Contextual Ambiguity

Polysemous proper nouns represent an inherent challenge:

- “*Apple released a new product*” → **ORG** (correct)
- “*I ate an apple*” → **O** (correct)

BiLSTM-CRF, lacking pre-trained world knowledge, resolves such cases based on surface frequency patterns. BERT’s attention mechanism captures the broader sentence context that disambiguates the intended meaning.

3.6 Role of Contextual Embeddings in NER

The BiLSTM-CRF generates token representations from a static embedding lookup table initialized randomly. Each word is mapped to a fixed vector regardless of its surrounding context. In contrast, BERT produces **dynamic, context-dependent representations**: the representation of the word “*bank*” in “*a river bank*” is fundamentally different from its representation in “*a central bank*”.

This contextual sensitivity is the primary driver of BERT’s recall advantage: it can identify entities in constructions that BiLSTM-CRF has not seen during training, because the pre-trained encoder has generalized over billions of tokens of text. Furthermore, BERT’s

subword tokenization handles rare or morphologically complex names (e.g., “*Schwarzenegger*”) more gracefully than a word-level vocabulary with an <UNK> fallback.

4 Question 3: Text Summarization

4.1 Problem Formulation and Dataset

Automatic text summarization aims to produce a concise, coherent, and informative synopsis of a longer source document. Two paradigms are evaluated:

- **Extractive summarization:** Selects and concatenates the most salient sentences from the source document verbatim.
- **Abstractive summarization:** Generates novel sentences that paraphrase and synthesize the source content.

The **CNN/DailyMail** dataset (version 3.0.0) was used, evaluating on the first 20 test samples. This dataset pairs news articles with human-written multi-sentence highlights, providing a realistic and challenging benchmark. Gold-standard references (“highlights”) serve as the evaluation target for all metrics.

4.2 Methods

4.2.1 Extractive: TextRank

TextRank is a graph-based ranking algorithm that models sentences as nodes and their pairwise semantic similarity (via TF-IDF cosine similarity) as weighted edges. The algorithm applies PageRank-style propagation to assign salience scores to sentences and selects the top-ranked sentences as the summary. We selected the **top 3 sentences** per article using the `sumy` library.

TextRank is completely unsupervised — it requires no training data and produces deterministic outputs. Its fundamental limitation is that it can only select sentences that already exist in the source, producing summaries that may be grammatically correct but informationally redundant or poorly connected.

4.2.2 Abstractive: BART

BART (Bidirectional and Auto-Regressive Transformers, `facebook/bart-large-cnn`) is a denoising autoencoder trained to reconstruct corrupted text. Its encoder is bidirectional (like BERT) and its decoder is autoregressive (like GPT). The `bart-large-cnn` checkpoint was specifically fine-tuned on the CNN/DailyMail dataset, making it the gold standard for this task.

Generation parameters:

- `max_length=130, min_length=30`

- `num_beams=4` (beam search)
- `length_penalty=2.0` (encourages longer summaries)
- Input truncated to 1024 tokens (BART’s context window limit)

4.3 Evaluation Metrics

Four complementary metrics were used:

- **ROUGE-1, ROUGE-2, ROUGE-L:** Measure unigram, bigram, and longest common subsequence overlap between the generated and reference summaries. Captures lexical fidelity.
- **BLEU:** Measures n-gram precision (modified) with brevity penalty. More sensitive to exact phrase matches.
- **METEOR:** Extends BLEU by incorporating stemming, synonymy, and paraphrase matching, making it more appropriate for abstractive evaluation.
- **BERTScore:** Computes soft token-level similarity using contextual BERT embeddings, capturing semantic equivalence beyond exact lexical matches.

4.4 Results

Table 6: Summarization performance on CNN/DailyMail test set (20 samples).

Metric	Extractive (TextRank)	Abstractive (BART)
ROUGE-1	0.226342	0.358884
ROUGE-2	0.064241	0.166513
ROUGE-L	0.156561	0.279672
BLEU	0.033939	0.130115
METEOR	0.288793	0.346544
BERTScore	0.852576	0.880121

4.4.1 Inter-Metric Relationships and Interpretation

The six metrics are not independent; they measure overlapping but distinct aspects of summary quality, and understanding their relationships is essential for correctly interpreting the results.

ROUGE-1 and ROUGE-2. ROUGE-1 counts unigram overlap between the generated summary and the reference, while ROUGE-2 counts bigram overlap. ROUGE-1 is a proxy for *vocabulary coverage* — does the summary mention the same concepts as the reference? ROUGE-2 additionally tests *local phrase fluency* — does the summary reproduce the same short phrases in the same order? Extractive models tend to score

competitively on ROUGE-1 because they copy verbatim sentences from the source, which naturally shares vocabulary with the reference. However, because extractive sentences are full-length and the reference highlights are concise, ROUGE-2 can be lower than expected: the bigram sequences of a long copied sentence often diverge from those of a short paraphrased reference. BART, by generating shorter, paraphrase-style summaries, may achieve comparable or higher ROUGE-2 despite lower ROUGE-1, because its compressed phrasing can more closely match the concise reference structure.

ROUGE-L. ROUGE-L measures the *longest common subsequence* (LCS) between prediction and reference, normalised by reference length. It is sensitive to global word-order preservation rather than local phrase matches. A summary that reproduces facts in the same narrative order as the reference — even using paraphrased phrasing — will score well on ROUGE-L. BART’s tendency to reorganise information (“The Palestinian Authority becomes the 123rd member...”) can yield higher ROUGE-L than TextRank if the selected sentences happen to be ordered differently from the reference highlights.

BLEU. BLEU computes modified n-gram precision (1- through 4-gram) with a brevity penalty. It is the strictest lexical metric: it rewards short, precise phrase matches and penalises redundantly long outputs. Extractive summaries are typically verbose (full sentences), which can trigger the brevity penalty only in the opposite direction — long summaries relative to short references reduce per-token precision. BART’s outputs, being condensed, often score better on BLEU than TextRank despite being abstractive, precisely because their length better matches the concise reference highlights.

METEOR. METEOR extends BLEU by aligning tokens based on exact match, stemmed match, and synonym match (via WordNet). This makes it the most linguistically aware lexical metric and the most appropriate for evaluating abstractive systems. For example, if BART generates “*jurisdiction over alleged crimes*” while the reference says “*authority over alleged offences*”, METEOR will partially reward the synonym pair (crimes/offences) and stemmed match, while BLEU would score zero for that phrase. We therefore expect METEOR to show a smaller gap — or even a reversal — between TextRank and BART compared to BLEU, because BART’s paraphrases are semantically close even when lexically distinct.

BERTScore. BERTScore computes the F1 of token-level cosine similarities between contextual BERT embeddings of the prediction and reference, bypassing surface-level lexical matching entirely. It is the only metric that can reward meaning-preserving paraphrases regardless of word choice. BERTScore is expected to be the most favourable metric for BART: its abstractive rewrites are semantically equivalent to the reference even when no single exact phrase is shared. Conversely, TextRank’s verbatim sentence copies may score lower on BERTScore when the selected sentences contain contextually irrelevant detail that dilutes the embedding similarity signal.

Overall metric convergence. When all metrics agree (both models rank the same), confidence in the conclusion is high. When they diverge — for example, TextRank wins on ROUGE-1 but BART wins on BERTScore — this signals the fundamental tension between *lexical faithfulness* and *semantic adequacy*. In summarization research, BERTScore and METEOR are now considered the more reliable proxies for human quality judgements, as they better tolerate the valid lexical variation introduced by abstractive generation.

4.5 Qualitative Analysis

Three examples from the CNN/DailyMail test set are examined below. For each, we analyse fluency, factual consistency, and information coverage across both methods.

4.5.1 Example 1 — Palestinian Authority / ICC Accession

Table 7: Example 1: Palestinian Authority joining the ICC.

Field	Text
Source (excerpt)	“The Palestinian Authority officially became the 123rd member of the International Criminal Court on Wednesday, a step that gives the court jurisdiction over alleged crimes in Palestinian territories.”
Gold Reference	“Membership gives the ICC jurisdiction over alleged crimes committed in Palestinian territories since last June. Israel and the United States opposed the move, which could open the door to war crimes investigations against Israelis.”
Extractive (TextRank)	“As Palestine formally becomes a State Party to the Rome Statute today, the world is also a step closer to ending a long era of impunity and injustice,” he said. Judge Kuniko Ozaki, a vice president of the ICC, said acceding to the treaty was just the first step for the Palestinians. “Palestine acquires all the rights as well as responsibilities that come with being a State Party to the Statute.”
Abstractive (BART)	“The Palestinian Authority becomes the 123rd member of the International Criminal Court. The move gives the court jurisdiction over alleged crimes in Palestinian territories. Israel and the United States opposed the Palestinians’ efforts to join the body.”

Fluency: Both outputs are grammatically correct. TextRank’s output is verbatim prose and reads naturally in isolation, but the transition between the three extracted sentences is abrupt — the attribution “he said” introduces a speaker without prior context. BART’s output is coherent and self-contained across all three sentences.

Information coverage: BART covers three key facts: (1) the membership number (123rd), (2) the jurisdictional consequence, and (3) the US/Israel opposition — exactly matching the reference highlights. TextRank focuses on the procedural and ceremonial dimension (ICC officials’ quotes) while omitting the politically salient US/Israel opposition entirely.

Factual consistency: No hallucinations are detectable in either output for this example. Both are factually grounded in the source article.

4.5.2 Example 2 — Miracle Survival Dog “Theia”

Table 8: Example 2: Stray dog Theia’s survival story.

Field	Text
Source (excerpt)	“A stray pooch in Washington State has used up at least three of her own after being hit by a car, apparently whacked on the head with a hammer in a misguided mercy killing and then buried in a field – only to survive.”
Gold Reference	“Theia, a bully breed mix, was apparently hit by a car, whacked with a hammer and buried in a field. ‘She’s a true miracle dog and she deserves a good life,’ says Sara Mellado, who is looking for a home for Theia.”
Extractive (TextRank)	A stray pooch in Washington State has used up at least three of her own after being hit by a car, apparently whacked on the head with a hammer and then buried in a field – only to survive. Four days after her apparent death, the dog managed to stagger to a nearby farm, dirt-covered and emaciated. She is dedicated to making sure Theia gets the medical attention she needs and wants to “make sure she gets placed in a family where this will never happen to her again!”
Abstractive (BART)	“Theia, a one-year-old bully breed mix, was hit by a car and buried in a field. Four days after her apparent death, the dog managed to stagger to a nearby farm. She suffered a dislocated jaw, leg injuries and a caved-in sinus cavity. She still requires surgery to help her breathe.”

Fluency: TextRank’s first sentence is exceptionally vivid and compelling — it was clearly the most salient lead sentence in the source. However, the third extracted sentence abruptly shifts to an unnamed “she” (Sara Mellado), creating a coreference gap. BART’s output is clean, structured, and reads as a natural news lead.

Information coverage: TextRank includes the emotional human-interest angle (“mir-

acle dog”, desire for adoption) but omits specific medical details. BART provides more clinical detail (dislocated jaw, surgery needed) at the cost of omitting the human adoption narrative present in the reference.

Factual consistency: BART introduces the detail “one-year-old” which, if not explicitly stated in the source excerpt provided, would constitute a minor hallucination — a characteristic risk of abstractive generation that TextRank is immune to by design.

4.5.3 Example 3 — Mohammad Javad Zarif Profile

Table 9: Example 3: Profile of Iranian Foreign Minister Mohammad Javad Zarif.

Field	Text
Source (excerpt)	“He is, of course, the Iranian foreign minister. He has been U.S. Secretary of State John Kerry’s opposite number in securing a breakthrough in nuclear discussions that could lead to an end to sanctions against Iran.”
Gold Reference	“Mohammad Javad Zarif has spent more time with John Kerry than any other foreign minister. He once participated in a takeover of the Iranian Consulate in San Francisco. The Iranian foreign minister tweets in English.”
Extractive (TextRank)	Zarif left the country in 1977, received his undergraduate degree from San Francisco State University in 1981, his master’s in international relations from the University of Denver in 1984 and his doctorate from the University of Denver in 1988. The first sentence of his official biography says Zarif was “born to a religious traditional family in Tehran in 1959.” [+ one further sentence about the Shah.]
Abstractive (BART)	“Mohammad Javad Zarif is the Iranian foreign minister. He has been John Kerry’s opposite number in securing a breakthrough in nuclear discussions. But there are some facts about him that are less well-known.”

Fluency: TextRank’s output is grammatically correct and reads as factual biography, but is entirely misaligned with the reference: it selected biographical background sentences rather than the most newsworthy facts. BART’s output is fluent and naturally structured as a news hook — though the third sentence (“But there are some facts about him that are less well-known”) is a meta-commentary rather than a substantive fact, indicating that the model partially reproduced the article’s rhetorical framing rather than its content.

Information coverage: This example exposes a fundamental weakness of TextRank on profile/listicle articles. The three most salient facts in the reference (Kerry relationship, consulate takeover, English tweets) are all in non-central paragraphs; PageRank assigned

them low salience because they are not reinforced by cross-sentence similarity. BART also fails to capture the consulate and Twitter facts, suggesting that even contextual models struggle with multi-focus biographical articles.

Factual consistency: Neither model introduces factual errors in this example. BART’s output, while shallow, is entirely entailed by the source. TextRank’s biographical dates are copied verbatim and therefore accurate.

4.6 Discussion: Extractive vs. Abstractive Trade-offs

Computational cost. TextRank is a fully unsupervised, graph-based algorithm requiring only TF-IDF vectorisation and PageRank propagation — operations that complete in milliseconds on CPU with no GPU requirement and no pre-trained model weights. BART-large-cnn has approximately 400M parameters; each inference pass requires loading the full model into GPU memory and running autoregressive beam-search decoding, which is orders of magnitude more expensive. For large-scale summarization pipelines (e.g., summarizing thousands of documents per hour), the computational asymmetry between the two approaches is practically significant.

Readability and fluency. Extractive summaries are guaranteed to be grammatically correct in isolation — they reproduce polished journalistic sentences verbatim. However, stitching together sentences from different paragraphs often produces abrupt transitions, dangling references (e.g., “he said” without antecedent), and redundancy. BART generates summaries as coherent, purpose-written paragraphs with natural connective flow, which readers consistently rate as more readable in human evaluation studies. The trade-off is that BART’s fluency is *generated* rather than *guaranteed*: in rare cases it can produce grammatically well-formed but semantically inconsistent sentences.

Faithfulness to the source. TextRank is *intrinsically faithful*: because it can only select sentences that appear verbatim in the source, it cannot introduce information not present in the original document. This makes it the safer choice in domains where factual accuracy is non-negotiable (legal, medical, scientific). BART is susceptible to *hallucinations* — generating plausible-sounding details (e.g., ages, statistics, names) that are not present in or are inconsistent with the source. Example 2 above illustrates a mild instance of this. Mitigating hallucination in abstractive models is an active research area involving faithfulness-constrained decoding, fact verification post-processing, and retrieval-augmented generation.

4.7 Discussion

The trade-off between extractive and abstractive approaches mirrors the broader tension in NLP between faithfulness and fluency:

- **TextRank** achieves high faithfulness (no hallucination by construction) and requires no training, but produces grammatically incoherent and redundant summaries.
- **BART** produces fluent, informative, and well-structured summaries but is susceptible to hallucinations and requires significant computational resources for inference. Its superior BERTScore reflects that even when exact word overlap is lower, the semantic content better matches the reference.

For applications requiring strict factual accuracy (e.g., medical or legal summarization), extractive methods or hallucination-mitigating post-processing pipelines are preferred. For general-purpose news summarization, BART’s abstractive capability provides a clear qualitative advantage.

5 Question 4: Machine Translation

5.1 Problem Formulation and Dataset

Machine Translation (MT) aims to automatically convert text from a source language to a target language while preserving meaning, fluency, and style. We evaluate English-to-German translation using two neural models on the **Multi30k** dataset ([bentrevett/multi30k](https://github.com/bentrevett/multi30k), first 30 test samples), a collection of image captions translated by professional translators.

5.2 Models

5.2.1 Model 1 – Seq2Seq + Attention: T5-Small (Legacy)

`google-t5/t5-small` is a text-to-text encoder-decoder transformer trained to reformulate all NLP tasks as text generation problems. It uses a prefix (“*translate English to German:*”) to condition generation on the desired task. T5-small has 60M parameters and was pre-trained on the C4 corpus. While T5 uses self-attention throughout, its pre-training was not specialized for translation, making it a “legacy” baseline in this context.

5.2.2 Model 2 – Modern NMT: Helsinki-NLP MarianMT (OPUS)

`Helsinki-NLP/opus-mt-en-de` is part of the OPUS-MT project, offering a suite of MarianMT models specifically trained and optimized for bilingual translation. This model was fine-tuned on large-scale parallel corpora (OPUS) specifically for English→German translation, making it the domain-optimized SOTA baseline for this task.

Both models use **beam search decoding** with `num_beams=4` and `max_length=100`.

5.3 Evaluation Metrics

- **BLEU**: Standard MT metric measuring n-gram precision with brevity penalty. Sensitive to exact phrase matches.

- **METEOR**: Incorporates synonym matching and stemming; more robust than BLEU for morphologically rich languages like German.
- **ChrF**: Character-level F-score measuring character n-gram overlap. Particularly well-suited for German, whose agglutinative morphology means compound word matches at the character level even when exact word matches fail (e.g., “*Hochschule*” partially matches “*Hochschulgebäude*”).
- **BERTScore**: Semantic similarity computed in German (`lang="de"`), capturing meaning-preserving paraphrases.

5.4 Results

Table 10: Machine translation performance (English → German) on Multi30k test set (30 samples).

Metric	Seq2Seq T5-Small (Legacy)	Transformer OPUS (MarianMT)
BLEU	0.290911	0.430028
METEOR	0.604018	0.732661
ChrF	56.423300	66.116915
BERTScore	0.880920	0.906823

5.4.1 What Each Metric Captures

The four evaluation metrics are deliberately chosen to cover complementary aspects of translation quality. Taken together, they form a multi-dimensional portrait of model behaviour; interpreted individually, each can be misleading.

BLEU — Lexical Precision with Brevity Penalty. BLEU computes modified n-gram precision at four granularities (1- through 4-gram) and multiplies the result by a brevity penalty that discourages pathologically short translations. It is the de-facto standard metric for MT comparison across publications, making it useful for situating results in the literature. However, BLEU has known weaknesses in the present experimental setting. German is a morphologically rich, agglutinative language: a single English word can correspond to a German compound or inflected form (e.g., “*a dog*” → “*ein Hund*” vs. “*des Hundes*” depending on case). A translation that is semantically and grammatically correct but uses a different case inflection than the reference will receive zero credit at the 4-gram level under BLEU. This systematic penalty on inflectional variants causes BLEU to underestimate the true quality of both models, but particularly of a general-purpose model like T5-small, which may produce valid alternative phrasings more often than the domain-specialised MarianMT.

METEOR — Synonym- and Stem-Aware Recall. METEOR computes a token-level F-score between prediction and reference after performing three tiers of alignment:

exact match, stemmed match, and synonym match (via WordNet). The inclusion of stemming is especially valuable for German, where the same lexical root produces many surface forms through case, gender, and number inflection (e.g., “*groß*”, “*großen*”, “*großer*” all stem to “*groß*”). A model that produces the correct root in the wrong inflection will receive partial credit under METEOR but zero credit under BLEU. We therefore expect METEOR scores to be higher relative to BLEU for both models, and the *gap between T5-small and MarianMT to narrow* under METEOR compared to BLEU, because METEOR rewards T5’s semantically adequate but morphologically imprecise outputs more fairly. If this pattern is observed in the results, it confirms that a portion of T5’s BLEU deficit is attributable to morphological mismatch rather than semantic error.

ChrF — Character-Level F-Score. ChrF computes precision and recall of character n-grams (by default, up to 6-grams) between the prediction and reference, then combines them into an F-score. The character-level granularity makes ChrF uniquely suited to German translation evaluation for two reasons. First, German compound nouns are written as single orthographic words (“*Hochschulgebäude*” = university building), meaning a model that correctly translates each constituent morpheme but merges or splits the compound slightly differently will receive partial character-level credit. Second, German inflectional suffixes (-en, -er, -em, -es) differ by only one or two characters from each other; ChrF rewards these near-misses where word-level metrics give zero. ChrF is therefore the most forgiving metric towards morphologically plausible errors and the most informative for understanding where a model’s German generation breaks down at the sub-word level. A substantially higher ChrF than BLEU for T5-small would indicate that the model is morphologically “in the right neighbourhood” even when its exact word forms do not match the reference.

BERTScore — Contextual Semantic Similarity. BERTScore computes the F1 of token-level cosine similarities between contextual multilingual BERT embeddings of the prediction and the reference (`lang="de"` ensures a German-specialised embedding space is used). It is the only metric evaluated here that operates entirely in semantic space, bypassing surface form. A translation that conveys the correct meaning using completely different vocabulary — synonyms, paraphrases, word reordering — will still score well on BERTScore. This makes it the most appropriate metric for capturing whether the *communicative intent* of the source sentence has been preserved, independent of how closely the model’s lexical choices match the specific reference translation. BERTScore is expected to show the smallest absolute gap between T5-small and MarianMT, because T5’s outputs, despite being lexically less precise, are likely to preserve overall sentence meaning. If BERTScore is disproportionately high for T5 relative to its BLEU score, this provides evidence that T5’s errors are primarily morphological and lexical rather than semantic.

Inter-metric convergence and divergence. When all four metrics rank MarianMT above T5-small, the conclusion that task-specialised pre-training produces superior translation quality is robust. The more interesting and informative case is when the metrics diverge: a large BLEU gap combined with a small BERTScore gap would indicate that MarianMT’s advantage is primarily lexical precision (correct word forms, domain vocabulary) rather than semantic understanding. Conversely, a large gap on ChrF would reveal morphological generation differences — MarianMT more consistently producing the correct German case endings and compound structures. Analysts should treat the full metric profile as a diagnostic fingerprint of each model’s strengths and failure modes rather than collapsing results to a single number.

5.5 Qualitative Example

Table 11: Qualitative translation comparison for Multi30k test sample (idx=0).

Field	Text
Source (EN)	A man in an orange hat starring at something.
Reference (DE)	Ein Mann mit einem orangefarbenen Hut, der etwas anstarrt.
T5-Small Output	Ein Mann in einem orangen Hut, der auf etwas dreht.
MarianMT Output	Ein Mann in einem orangenen Hut, mit etwas.

Lexical precision. The reference uses the precise compound adjective *orangefarbenen* (“orange-coloured”), the standard literary form for colour attribution in German. T5-small produces *orangen* and MarianMT produces *orangen* — both are informal but grammatically plausible inflected forms of *orange* used colloquially. Neither matches the reference exactly, explaining a low BLEU score for both models on this token. However, ChrF will award partial credit to both, since the character sequences *orange-* are shared across all three forms. This is precisely the scenario where ChrF is more informative than BLEU for German evaluation.

Semantic fidelity and the critical verb error. The most significant error in this example is T5-small’s mistranslation of the verb. The source verb “*starring at*” (gazing intently) should translate to *anstarrt* or *schaut auf*. T5-small produces *dreht* (“turns” or “rotates”), which is semantically unrelated — a clear lexical substitution error rather than a morphological one. This is a case where BERTScore will also penalise T5, since the sentence meaning is materially altered: a man *turning* something is a fundamentally different action from a man *staring at* something. This single verb error illustrates that

T5-small’s weakness is not confined to morphology; it extends to domain-specific verb phrase translation in the image caption domain.

Information coverage and truncation. MarianMT’s output — “*Ein Mann in einem orangenen Hut, mit etwas*” — omits the verb phrase entirely, producing a grammatically incomplete sentence (a noun phrase followed by a bare prepositional phrase). The participial relative clause that should encode the staring action is absent. This truncation error means MarianMT fails to convey the core predicate of the sentence, which will depress its METEOR and BERTScore despite its otherwise correct noun phrase structure. This is a notable failure for a supposedly task-specialised model and may reflect beam search finding a higher-probability short completion than the correct full translation on this particular example.

Summary. Neither model produces a high-quality translation for this sentence. T5-small attempts a complete sentence but introduces a critical semantic verb error (*dreht* vs. *anstarrt*). MarianMT produces a morphologically plausible noun phrase but truncates the output, omitting the predicate entirely. This example demonstrates that metric averages over 30 samples can mask individual failure modes: a model with higher average BLEU may still produce severely degraded outputs on specific sentences, and qualitative inspection remains an irreplaceable complement to aggregate quantitative evaluation.

5.6 Discussion: Architecture Trade-offs

Task specialisation vs. generality. MarianMT (opus-mt-en-de) was trained exclusively on large-scale English–German parallel corpora from the OPUS collection, making every parameter in the model optimised for this single translation direction. T5-small, by contrast, is a generalist text-to-text model conditioned on a task prefix; its parameters must simultaneously support summarization, question answering, translation across many language pairs, and other NLP tasks. This multitask compression inevitably sacrifices specialised translation quality. The consistent metric advantage of MarianMT across all four evaluation dimensions is a direct empirical consequence of this specialisation gap.

Domain alignment. The Multi30k test set consists of *image captions* — short, visually grounded sentences describing concrete scenes (“*A dog runs across the grass*”). MarianMT’s OPUS training data includes large parallel news, legal, and web corpora but was calibrated specifically for En→De, meaning its vocabulary and phrase table are likely to include common caption-domain entities (animals, colours, spatial prepositions, outdoor settings). T5-small’s En→De training signal is embedded within a much broader multilingual pre-training mix, and its tokenizer is not optimised for German morphology. Caption-domain sentences — which rely heavily on correct gender agreement for concrete nouns (*Hund/der*, *Frau/die*) — amplify the morphological weaknesses of a general-purpose tokenizer.

Morphological handling. German’s four-case inflection system and compound noun formation represent the most significant structural challenge for English-to-German translation. BLEU penalises any surface mismatch regardless of semantic validity; ChrF and METEOR partially reward morphologically proximate outputs. The observation that ChrF scores are substantially higher than BLEU for both models (expected given German morphology) confirms that errors are concentrated at the inflectional level rather than at the level of lexical selection or syntactic structure. MarianMT’s domain-specialised BPE tokenizer, trained on En–De parallel data, produces subword segmentations that more faithfully encode German morphological boundaries, giving it a systematic advantage on all four metrics.

Computational cost and deployment. T5-small (60M parameters) is considerably more lightweight than MarianMT-base (approximately 74M parameters for the En→De pair), but this size difference alone does not account for the quality gap. The more important factor is the inference overhead introduced by T5’s task prefix mechanism and its generalist vocabulary (32,128 sentencepiece tokens covering multiple languages and tasks), which increases the token budget and reduces generation efficiency compared to MarianMT’s bilingual vocabulary. For production En→De translation systems, MarianMT offers both superior quality and more efficient specialised inference.

Conclusion. The four metrics jointly reveal that MarianMT’s advantage over T5-small is multi-dimensional: it produces more lexically precise translations (BLEU), handles German morphology more accurately (ChrF), better preserves stem-level meaning (METEOR), and maintains equivalent or superior semantic fidelity (BERTScore). T5-small remains a viable option in resource-constrained or multi-task settings where a single model must cover many languages and tasks, but for any production deployment of a single translation direction, a task-specialised model such as MarianMT will consistently outperform a general-purpose architecture of comparable size.

5.7 Problem Formulation and Dataset

Language modeling is the task of assigning a probability to a sequence of tokens, or equivalently, predicting the next token given all preceding tokens. It is one of the most fundamental problems in NLP and underpins virtually all modern generative systems. In this task we compare two paradigms:

- **Statistical N-gram model (Bigram):** Estimates the probability of each token given only the immediately preceding token, using maximum-likelihood estimation over corpus counts.
- **Neural Transformer model (GPT-2):** A 117M-parameter autoregressive decoder pre-trained on WebText, generating each token conditioned on its full left context via masked self-attention.

The **WikiText-2** dataset (`wikitext-2-raw-v1`) was selected as the benchmark corpus. It consists of verified, high-quality English Wikipedia articles spanning approximately 2M training tokens and 245K test tokens. Its key advantages for language modeling research are:

- **Linguistic diversity:** Encyclopedic text spans a broad range of topics, domains, and sentence structures, challenging a model’s ability to generalise.
- **Minimal noise:** Unlike web-scraped corpora, WikiText-2 has been cleaned of broken markup, making perplexity scores more reliable and reproducible.
- **Benchmark status:** WikiText-2 is a canonical language modeling benchmark, allowing comparison with published results.
- **Manageable size:** The corpus is small enough to train an N-gram model entirely in memory while still being representative.

All empty lines were removed and the remaining lines were concatenated into a single token stream using whitespace splitting, yielding a flat sequence of whitespace-delimited tokens for both training and evaluation.

5.8 Model Architectures

5.8.1 Model 1 – Statistical: Bigram Language Model

A bigram language model factorises the joint probability of a token sequence using the first-order Markov assumption:

$$P(w_1, w_2, \dots, w_T) \approx \prod_{t=2}^T P(w_t \mid w_{t-1})$$

Each conditional probability is estimated by maximum-likelihood estimation (MLE) over the training corpus:

$$P(w_t \mid w_{t-1}) = \frac{C(w_{t-1}, w_t)}{C(w_{t-1})}$$

where $C(\cdot)$ denotes the raw frequency count. The model is stored as a nested dictionary mapping each context word w_{t-1} to a probability distribution over successor words.

Unknown word handling: For unseen bigram pairs (w_{t-1}, w_t) encountered during test-time evaluation, a back-off probability of $\frac{1}{|V_{\text{train}}|}$ is assigned, where $|V_{\text{train}}|$ is the size of the training vocabulary. This is a minimal form of smoothing sufficient for perplexity estimation on a domain-matched test set.

Text generation: Given a seed word, the model samples the next word proportionally to the bigram probability distribution. This process is repeated autoregressively for up to 15 steps or until the current word has no recorded successors in the model.

5.8.2 Model 2 – Neural: GPT-2

GPT-2 (`gpt2`, `openai-gpt2`) is a transformer decoder stack with 12 layers, 12 attention heads, and a model dimension of 768, totalling approximately 117M parameters. It was pre-trained on WebText — a large, diverse corpus of web pages — using the standard autoregressive language modeling objective:

$$\mathcal{L} = - \sum_t \log P(w_t \mid w_1, \dots, w_{t-1})$$

GPT-2 processes sequences using masked (causal) multi-head self-attention, allowing every token to attend to all previous tokens in a single forward pass. Its byte-pair encoding (BPE) tokenizer with a 50,257-token vocabulary handles rare and out-of-vocabulary words gracefully through subword segmentation.

Text generation configuration:

- `max_length=40`
- `do_sample=True`, `top_k=50` (top- k sampling for naturalness)
- `no_repeat_ngram_size=2` (prevents immediate repetition)
- `pad_token_id=tokenizer.eos_token_id`

5.9 Evaluation: Perplexity

Perplexity (PPL) is the standard intrinsic evaluation metric for language models. It measures how surprised a model is by an unseen test sequence:

$$\text{PPL} = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(w_i \mid w_{<i}) \right)$$

Lower perplexity indicates that the model assigns higher probability to the observed text, reflecting better predictive calibration. A perplexity of k means the model is, on average, as uncertain as if it were choosing uniformly among k equally likely options at each step.

Bigram perplexity was computed directly from bigram conditional probabilities using base-2 logarithms over a 500-token test window (`test_tokens[100:600]`). **GPT-2 perplexity** was computed from the model's cross-entropy loss on the same text segment, truncated to GPT-2's 1,024-token context window.

5.10 Results

Table 12: Language model perplexity on a 500-token WikiText-2 test segment.

Model	Perplexity	Parameters
Bigram (MLE)	2138.93	$\sim V ^2$ counts
GPT-2	23.69	117M

The results demonstrate a dramatic gap between the two models. The Bigram model’s perplexity is expected to be very high (typically in the hundreds to thousands), reflecting its fundamental limitation: it can only condition on a single preceding word, causing it to assign near-zero probability to grammatically or semantically unexpected but valid continuations. GPT-2’s perplexity is expected to be substantially lower, thanks to its ability to capture long-range dependencies across its full 1,024-token context window and its rich pre-trained linguistic knowledge.

5.11 Qualitative Text Generation Analysis

Table 13: Sample text generated by each model. Seed for Bigram: “*The*”; prompt for GPT-2: “*The scientific discovery of*”.

Model	Generated Text
Bigram (MLE)	<i>The Labour and Isidore in thick atmosphere , such attempt to the dispute between the time</i>
GPT-2	<i>The scientific discovery of the existence of light is in the very nature of natural laws, which I have already said. But inasmuch as it is the most important part of all, but owing to</i>

Bigram generation analysis: The Bigram model produces locally plausible word pairs — adjacent tokens are statistically co-occurring in the training corpus — but the global sentence structure is incoherent. There is no mechanism to maintain topic, syntactic structure, or argument consistency beyond a single token of history. Common failure modes include abrupt topic shifts between consecutive bigrams, grammatical agreement violations (e.g., number or tense mismatches), and premature termination when a generated word lacks any recorded successor.

GPT-2 generation analysis: GPT-2 produces grammatically well-formed, topically coherent text that reads as natural English prose. The `no_repeat_ngram_size=2` parameter prevents degenerate repetition loops, and `top_k=50` sampling introduces lexical diversity without degrading fluency. The use of full causal self-attention allows GPT-2 to maintain syntactic agreement and thematic consistency across the entire generated span.

5.12 Discussion: Model Behavior and Limitations

5.12.1 Bigram Model

The Bigram model embodies the n-gram paradigm’s core trade-off: its extreme simplicity makes it fast, fully interpretable, and trainable on any corpus without GPU resources, but its expressive capacity is fundamentally bounded by the Markov order. Key limitations observed include:

- **Data sparsity:** Even on a moderately sized corpus like WikiText-2, the vast majority of possible bigram pairs are never observed, leading to frequent fallback to the uniform unknown-word probability. This inflates perplexity.
- **No long-range dependencies:** Sentence-level phenomena like subject-verb agreement, pronoun coreference, and topical coherence require conditioning on tokens far outside the bigram window.
- **No semantic generalisation:** Synonym substitution is invisible to the model; *“large”* and *“big”* are entirely independent vocabulary entries with no shared distributional properties.

5.12.2 GPT-2

GPT-2 addresses all of the above limitations through its transformer architecture and large-scale pre-training, but introduces its own distinct limitations:

- **Computational cost:** Inference requires loading a 117M-parameter model and performing a full forward pass through 12 transformer layers, making it several orders of magnitude more expensive than a bigram lookup.
- **Domain mismatch:** GPT-2 was pre-trained on WebText (informal web content), while WikiText-2 is formal encyclopedic text. This domain gap may inflate GPT-2’s perplexity relative to a model fine-tuned on Wikipedia.
- **Hallucination:** GPT-2 can produce confident-sounding but factually incorrect statements, a limitation irrelevant to the Bigram model which can only reproduce observed n-grams.
- **Context window:** While 1,024 tokens is far beyond any n-gram model’s scope, it still imposes a hard upper bound on long-document coherence.

5.12.3 Summary

Table 14: Qualitative comparison of the two language modeling paradigms.

Dimension	Bigram (MLE)	GPT-2 (Transformer)
Context window	1 token	Up to 1,024 tokens
Perplexity	Very high	Low
Fluency	Poor	High
Coherence	None	Strong
Interpretability	High (counts)	Very Low
Training cost	Negligible	Very High (pre-training)
Inference speed	Instant	Moderate (GPU required)
Generalisation	None	Strong (pre-trained)

The contrast between the Bigram and GPT-2 models encapsulates the broader arc of progress in language modeling: from hand-designed statistical assumptions and sparse count tables to end-to-end learned, massively parameterised neural systems that implicitly acquire syntax, semantics, and world knowledge from raw text. The perplexity gap between the two models is not merely a reflection of scale, but of architectural expressiveness — GPT-2’s attention mechanism can condition on arbitrarily distant context, something no finite-order n-gram model can achieve regardless of the amount of training data.

6 Conclusion

This midterm examined the representational hierarchy of NLP models — from sparse statistical features to dense sequential encoders to large pre-trained transformers — across four distinct tasks. The following overarching conclusions emerge:

1. **Contextual representations consistently dominate.** Across all tasks, transformer-based models (DistilBERT, BERT, BART, MarianMT) substantially outperformed classical and recurrent architectures, primarily because of the breadth of linguistic knowledge encoded during large-scale pre-training.
2. **Task specialization matters.** Within the transformer family, models fine-tuned on task-specific data (MarianMT for En→De, BART-large-cnn for summarization) significantly outperform general-purpose models (T5-small), emphasizing that architecture alone does not determine performance.
3. **Pre-training bridges the data scarcity gap.** BiLSTM models trained from scratch for only 2 epochs cannot match the knowledge acquired by transformers during months of large-scale pre-training, even on comparable model sizes.

4. **Performance and interpretability trade inversely.** TF-IDF + Logistic Regression remains the most interpretable and computationally efficient option; its performance on sentiment tasks is surprisingly competitive. For production systems requiring explainability (e.g., in healthcare or legal NLP), this trade-off is non-trivial.
5. **Evaluation requires multiple metrics.** No single metric captures all aspects of NLP quality. The combination of lexical (ROUGE, BLEU), morphological (ChrF), and semantic (BERTScore, METEOR) metrics provides a more complete picture of model behavior.

Reproducibility Statement

All experiments were conducted with a global random seed of 42. The environment uses Python 3.10+, PyTorch 2.x, and Hugging Face `transformers` 4.x. All hyperparameters, dataset splits, and model checkpoints are as described in this report. A Colab notebook containing fully runnable code is provided at the repository link above.

Key hyperparameters summary:

- All neural models: 2 training epochs, batch size 32 (Q1) / 16 (Q2), Adam optimizer.
- TF-IDF: 5,000 features, lowercase normalization.
- BiLSTM: `embed_dim=128`, `hidden_dim=64`, `max_len=64`.
- DistilBERT / BERT: `max_length=128` (Q1), cased (Q2).
- BART: `max_length=130`, `min_length=30`, `num_beams=4`.
- MarianMT / T5: `max_length=100`, `num_beams=4`.

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